

QUERY LANGUAGES

*Keyword-Based Querying · Pattern Matching · Structural Queries · Query
Protocols*

Single-Word Queries

python

ranked by relevance

Python (programming language) — official site



Python (snake) — species overview



Python Software Foundation



One word in, a ranked list out — the simplest query language there is.

Phrase Queries

"state bank of india"

✓ "...loans from the **state bank of india**
last year"

✗ "...the state has a bank, but india's..."

Same four words, different order and spacing — only the exact adjacent sequence matches.

Proximity Queries

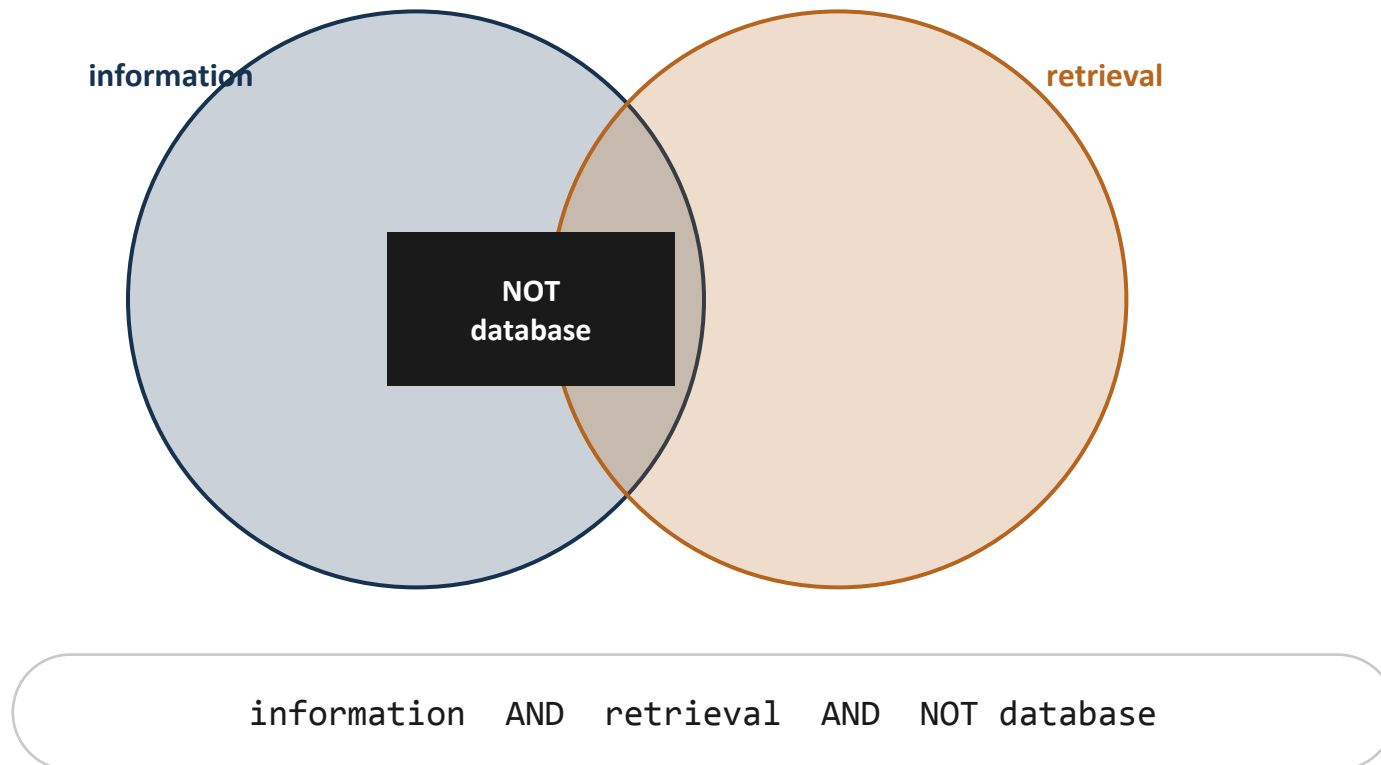
“...a survey of **information** systems used for **retrieval** in Indian libraries...”



within 5 words

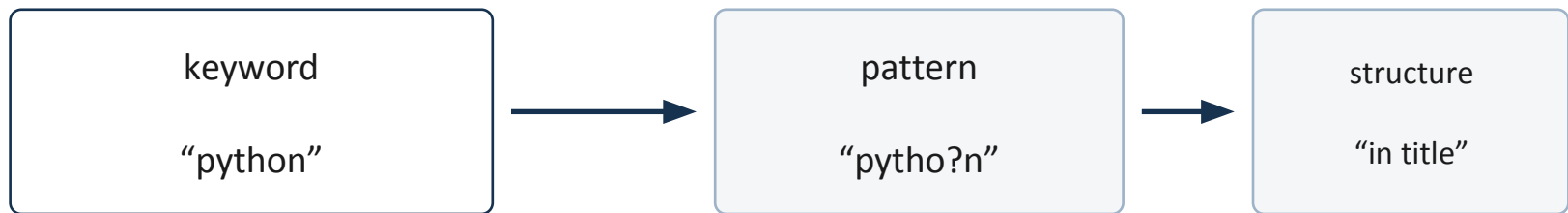
A looser phrase query — a maximum gap allowed between terms, order sometimes relaxed too.

Boolean Queries



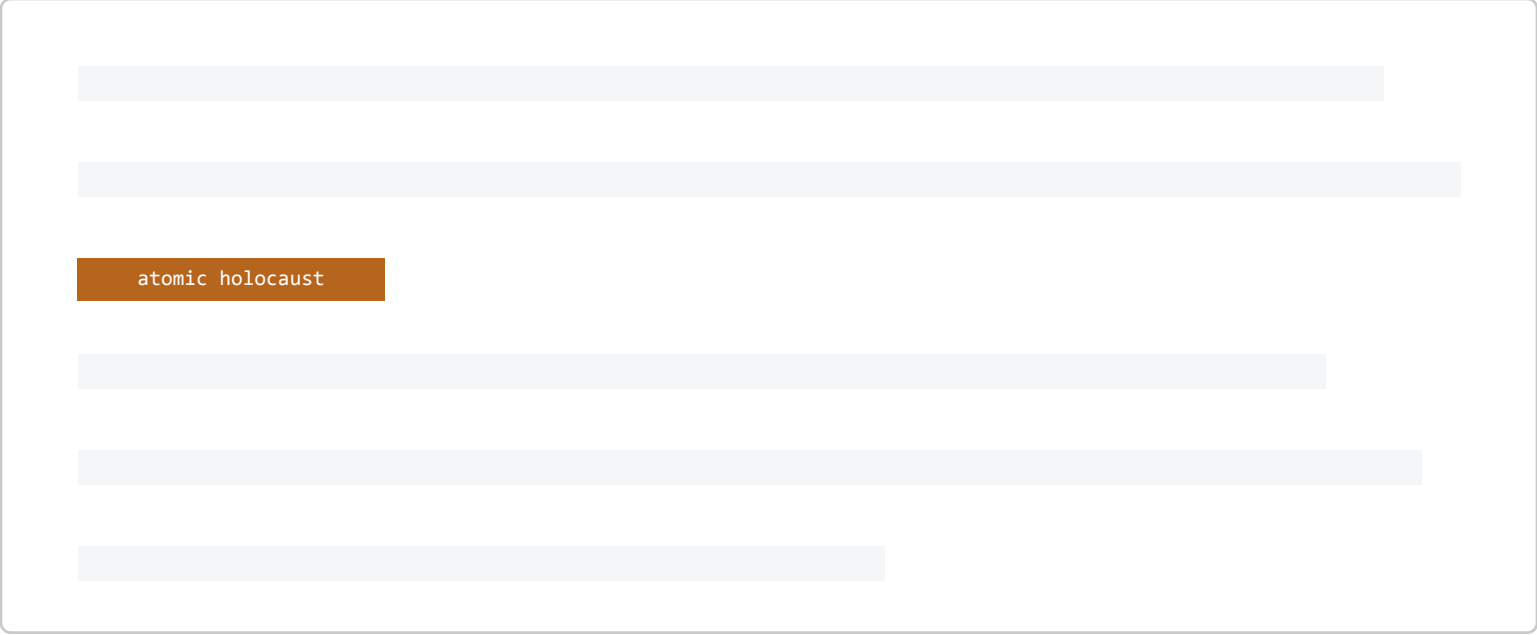
All-or-nothing matching — no partial credit, no ranking, exactly like the Boolean retrieval model.

From ‘A Word’ to ‘Exactly This, Exactly Here’



Each step adds one more precise condition on top of the last.

Pattern Matching



The diagram shows a large light gray rounded rectangle containing six horizontal bars of varying lengths, also in light gray. The third bar from the top is highlighted in a solid brown color and contains the text 'atomic holocaust' in a white, lowercase, monospaced font. This visualizes a specific pattern being matched within a larger text block.

atomic holocaust

a set of exact syntactic features that must literally occur in the text

A pattern isn't a meaning — it's a literal shape the characters must match.

Prefix Matching

comput |

comput er
comput ation
comput ing
comput ed

Real life: this is exactly what happens the instant you start typing into any search box.

Substring Matching

computer

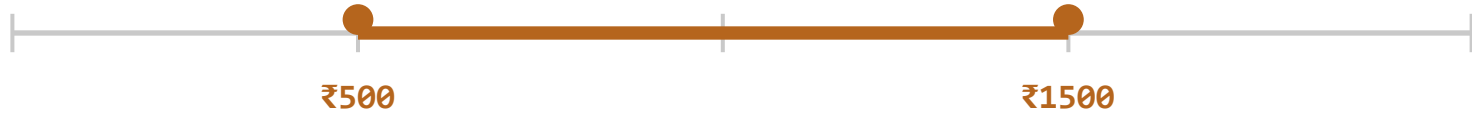
input

output

reputation

The pattern can start anywhere inside the word — powerful, but costlier to index than a prefix.

Range Queries



all prices between ₹500 and ₹1500

Anything with a natural order — price, date, page number — can be searched as a range.

Allowing Errors

pyhton retrieval systm

Did you mean: *python retrieval system*

Matches allowed within a small edit distance — insert, delete, or substitute a few characters.

Regular Expressions

```
$ grep 'colou?r' report.txt
```

```
> the wall was painted a bright color
```

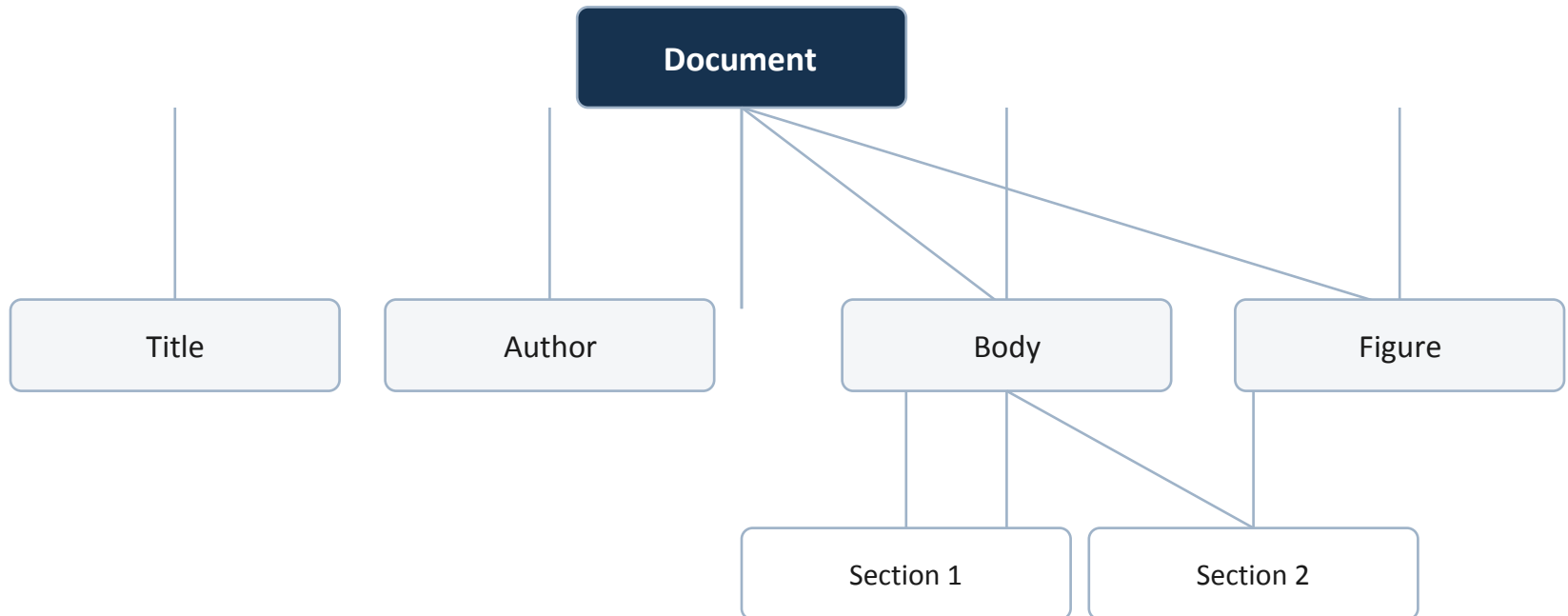
```
> British spelling uses colour
```

```
> no match: colouring
```

one query, two spellings — the ? makes the 'u' optional

Anything you can do with grep or 'Find with regex' in an editor is this exact idea.

Structural Queries



The same structure you already met in Structured Text Retrieval — now expressed as a query.

Structural Queries in Real Life

Advanced Search

find pages with... in the title

`intitle: python`

find pages with... in the domain

`site: mu.ac.in`

find pages of this file type

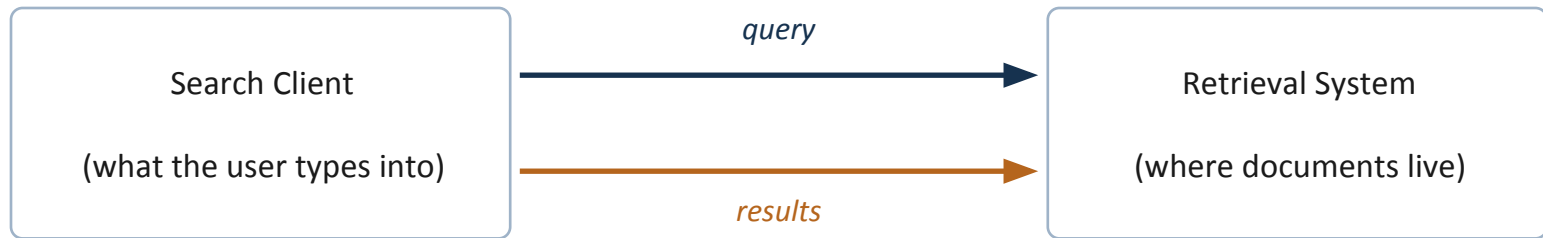
`filetype: pdf`

find pages with this exact phrase

`"information retrieval"`

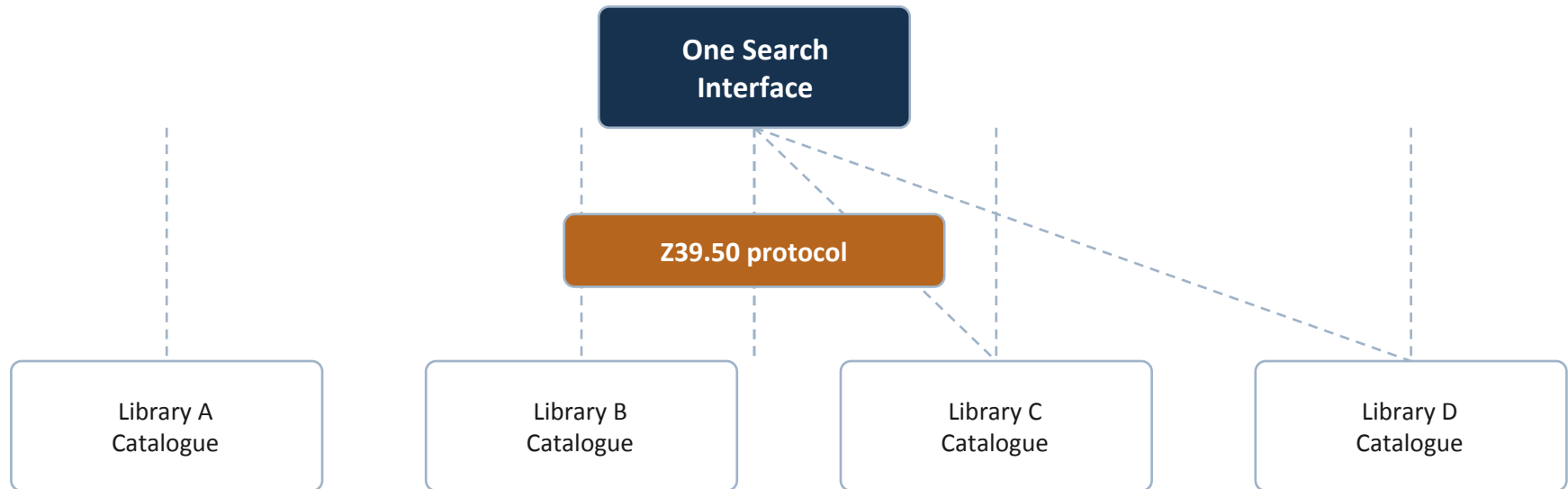
Every field-restricted search box — library databases included — is a structural query underneath.

Query Protocols



A query language is what you say. A query protocol is how it actually gets sent and answered.

Query Protocols in Real Life



One query, sent once, silently answered by many independent catalogues — the ancestor of federated search.